Volleyball Group Activity Recognition and Rally Prediction

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Abstract

This project presents a deep learning approach for volleyball activity recognition and rally prediction that analyzes volleyball rallies from past matches by predicting the success probabilities of rallies for the two teams. This will be useful for coaches to analyze the individual and group performance of the team in previous matches and provide feedback to the team. We frame this as a multi-task learning problem where individual player actions, team activities, and match outcomes must be predicted simultaneously. Our approach demonstrates that combining individual action recognition with spatial relationship modeling through graph networks significantly improves both group activity classification and rally prediction.

1 Introduction

Volleyball is a team sport characterized by intricate coordination patterns in which individual player actions combine to form group activities to build strategies. The ability to recognize these patterns from videos of previous matches can help coaches analyze team performance and build strategies to improve future performance.

The task of activity recognition in volleyball requires identification of both individual player actions (such as spiking or blocking) and collective team activities (such as left-pass or right-spike). Extending this to rally prediction — determining which team is likely to win the current exchange — introduces a higher-level understanding component that integrates group dynamics and game knowledge.

Previous work in sports video analysis has focused primarily on tracking players or recognizing isolated actions. More recent approaches have begun to model group activities, but often treat the problem as a simple classification without modeling the underlying relationships between players or predicting future outcomes.

Our work addresses these limitations through a multi-task approach that:

1. Uses a hierarchical model to recognize individual actions and group activities simultaneously
2. Incorporates graph-based representations to capture spatial relationships between players
3. Integrates temporal information through recurrent neural networks to model action sequences
4. Combines neural predictions with volleyball-specific heuristics to predict rally outcomes

Our model demonstrates improvements over existing methods in both action recognition and group activity classification, while introducing a new metric for rally prediction accuracy.

This approach represents a step toward AI systems that can understand not only what is happening in team sports, but also why certain actions are taking place and what their likely outcomes will be - capabilities that would be valuable for automated coaching and enhanced broadcasts in the future.

2 Dataset

The Volleyball Dataset is a benchmark for activity recognition in team sports, featuring annotated video clips of volleyball games. Each frame includes bounding boxes for up to 12 players (6 per team), with detailed labels for both individual actions and group activities.

Individual actions are categorized into nine classes: waiting, setting, digging, falling, spiking, blocking, jumping, moving and standing. Group activities are labeled in eight categories that capture team coordination, such as left/right pass, set, spike, and winpoint.

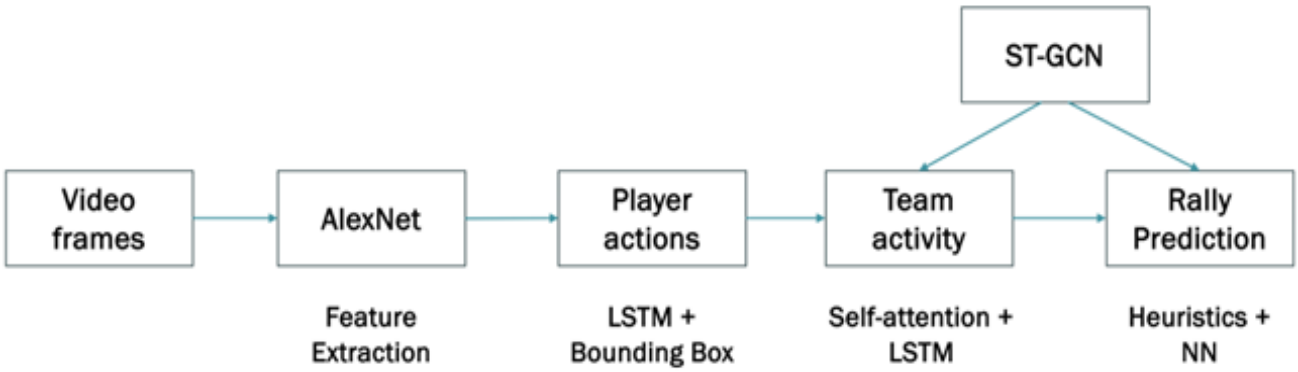


Figure 1: Model Flowchart

A major challenge in this dataset is class imbalance - standing and moving dominate, while actions like jumping and spiking are rare. This reflects real gameplay but complicates model training. The dataset shows extreme imbalance with standing (38,696 instances) and moving (5,121) dominating, while other actions appear less frequently: waiting (3,601), blocking (2,458), digging (2,333), setting (1,332), spiking (1,216), falling (1,241), and jumping (341).

To address this imbalance, we implement class weighting strategies during training, assigning higher weights to underrepresented classes. Additionally, we apply data augmentation techniques to increase the representation of minority classes. The dataset is split by video IDs into training, validation, and test sets. Some videos are high-resolution and require special preprocessing.

During preprocessing, we extract 10-frame sequences (5 frames before and 5 after the key frame) to capture the temporal context of each action. Each player is represented by a bounding box, and in cases where fewer than 12 players are detected, we pad with placeholder values to maintain consistent input dimensions for our neural network.

3 Model

Our system implements a hierarchical deep learning architecture as shown in Figure 1 that integrates multiple neural network components to model the complex dynamics of volleyball gameplay. The architecture consists of four primary components: a feature extractor, a person action classifier, a group activity classifier, and a rally predictor.

3.1 Feature Extraction

For visual feature extraction, we employ a pre-trained AlexNet CNN backbone to process each frame. The network's final fully connected layer outputs a 4096-dimensional feature vector for each frame. To maintain computational efficiency while leveraging pre-trained weights, we selectively freeze the early convolutional layers (layers 0-4) while fine-tuning the later layers (layers 5-12) during training.

3.2 Person Action Classifier

The Person Action Classifier processes individual player actions through a temporal sequence. For each player, we concatenate the extracted visual features (4096-d) with their normalized bounding box coordinates (4-d) to create a combined representation of dimension 4100. This input sequence is fed into a two-layer bidirectional LSTM network with 512 hidden units, which models the temporal dynamics of each player's movements across the 10-frame sequence.

The output from the LSTM is processed through batch normalization across the player dimension (batchSize $\times$ 12) and dropout (rate = 0.6) before passing through a fully connected layer that projects to 9 action classes. To address class imbalance, we implement a Focal Loss function with gamma = 1.0 and class-specific alpha weights derived from the inverse frequency of each action class. The weights are further adjusted to assign higher importance to dynamic actions (spiking = 1.5x, blocking = 1.5x etc.).

3.3 Group Activity Classifier

The Group Activity Classifier builds upon the individual player representations to model team-based activities. We implement a self-attention mechanism defined as:

$$Q = W_q * X \tag{1}$$

$$K = W_k * X \tag{2}$$

$$V = W_v * X \tag{3}$$

$$Attention(Q, K, V) = softmax(QK^T/sqrt(d_k)) * V \tag{4}$$

where X represents player features, Q, K, V are the query, key, and value matrices respectively, W_q, W_k, W_v are the corresponding learnable parameter

matrices, T is the transpose, and d_k is the dimension used for scaling.

This attention mechanism computes pairwise interactions between all players, generating attention scores that capture the relative importance of each player for the current play.

The attention-weighted features are processed by an LSTM layer with 512 hidden units that treats the 12 players as a sequence, modeling the group dynamics. The final hidden state is normalized (BatchNorm1d) and regularized (Dropout = 0.6) before classification through a fully connected layer to predict one of the eight group activity categories.

3.4 Spatio-Temporal Graph Convolutional Network (STGCN)

To explicitly model spatial and temporal relationships between players, we implement a Spatio-Temporal Graph Convolutional Network (STGCN). The STGCN operates on a graph structure where players are represented as nodes N, and edges E are dynamically determined based on Euclidean distances between player positions:

$$A_{ij} = \begin{cases} 1 & \text{if } ||p_i - p_j|| < threshold \\ 0 & \text{otherwise} \end{cases}$$

where p_i represents the center coordinates of player i's bounding box, and the threshold is set to 150 pixels. Self-loops are added to ensure information retention.

The STGCN consists of two layers with the following operations:

1. Graph convolution:
$$X' = D^{(-1/2)}AD^{(-1/2)}XW \tag{5}$$
2. Batch normalization in the feature dimension
3. ReLU activation: $X' = \sigma(X')$

   where D is the degree matrix of A, X is the input feature matrix, W is a learnable weight matrix, and σ is the ReLU activation function

This graph-based approach allows the model to learn teamwork patterns by enabling information exchange between nearby players (like setters and spikers), capturing critical volleyball formations and tactical dependencies that would be missed by analyzing players in isolation.

3.5 Rally Prediction Component

The Rally Prediction component combines neural network outputs with domain knowledge to estimate which team is likely to win the current rally, formulated as:

1. **Neural approach:** The player features from the last timestep are flattened (12×512) and processed through a fully connected network with architecture [6144, 512, 2] with ReLU activation and dropout (rate = 0.3), producing binary probabilities for left and right team victory.
2. **Heuristic approach:** A weighted scoring function assigns predefined values to different action types:
$$\text{Score} = \left(\sum_{i \in \text{team}} \text{weight}(\text{action}_i) \right) + \text{position_score}$$
   where action weights are {spiking: 1.5, setting: 1.0, blocking: 0.75, digging: 0.75, standing: 0.0, moving: 0.0, waiting: 0.0, jumping: 0.0, falling: 0.0}, and position_score (0.5 or 0.0) is determined by team proximity to the net.

During training, we employ two specialized loss functions:

1. **Margin Ranking Loss:**
$$L_{rank} = \max(0, -y(P_l - P_r) + \text{margin}) \tag{6}$$
   where $y \in \{-1, 1\}$ indicates the target team (1 for left team, -1 for right team), P_l and P_r are the predicted probabilities for left and right team victory respectively, and margin = 0.2 is a hyperparameter that enforces a minimum difference between the probabilities.
2. **MSE Loss between neural and heuristic predictions:**
$$L_{mse} = \|P_{neural} - P_{heuristic}\|^2 \tag{7}$$
   where P_{neural} represents the neural network's predicted probability distribution, $P_{heuristic}$ represents the heuristic-based probability distribution.

3.6 Training Strategy

We train the model end-to-end using a weighted multi-task loss function:

$$\begin{aligned} L_{total} = 2.5 * L_{person} + 2.5 * L_{group} + \\ 1.5 * L_{rank} + 1.0 * L_{mse} \end{aligned} \tag{8}$$

For optimization, we employ the AdamW optimizer with an initial learning rate of 0.0001, betas of (0.9, 0.999), and weight decay of $5e^{-3}$. The weight decay regularization helps prevent overfitting, particularly important given the relatively small dataset size.

To address class imbalance, we used separate loss functions for each task: Focal Loss (gamma=1.0) with class-specific alpha weights for person action classification, CrossEntropyLoss with group-activity weights for group classification, MarginRankingLoss (margin=0.2) for rally ranking, and MSELoss for rally heuristic consistency.

To improve training stability and performance, we implement:

- Gradient accumulation (steps = 2) for an effective batch size of 32
- Mixed-precision training using PyTorch's amp module
- Cosine annealing learning rate scheduling ($T_0 = 12$, $\eta_{min} = 5e^{-7}$)
- Gradient clipping ($max_norm = 1.0$)
- Model checkpointing based on validation group activity accuracy

The integrated architecture enables comprehensive volleyball activity understanding, from individual player actions to team coordination patterns and ultimately to rally outcome prediction.

4 Results

We evaluated our model on the Volleyball Dataset test set comprising 1,337 samples across several metrics to assess performance on individual action recognition, group activity classification, and rally outcome prediction tasks.

4.1 Individual Action Recognition

The model achieved an overall accuracy of 69.25% for individual player action classification. However,

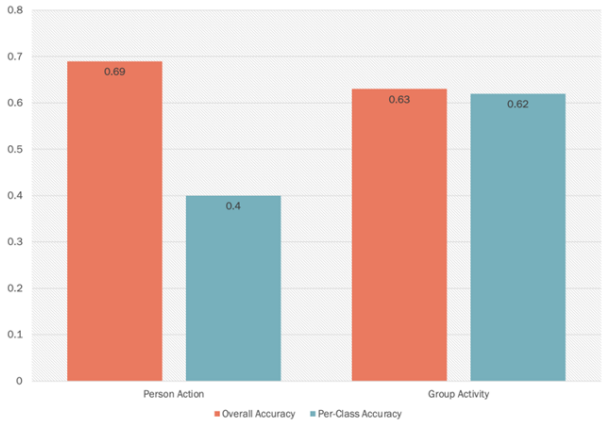


Figure 2: Overall vs Per-class Accuracy for Person Action Recognition and Group Activity Classification

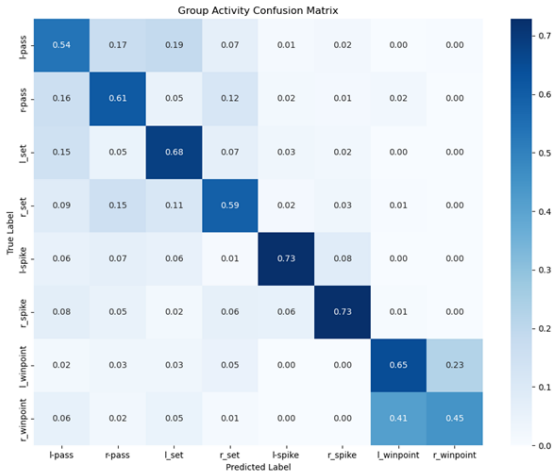


Figure 3: Confusion matrix for group activity classification across eight volleyball activities

when examining per-class accuracy, the average drops to 40.17%, indicating significant performance variation across action categories. This disparity highlights the challenge of recognizing less frequent or more dynamic actions in volleyball gameplay.

The performance gap between overall and per-class accuracy as shown in Figure 2 suggests that the model performs well on common actions that dominate the dataset but struggles with minority classes despite our implementation of class-weighted loss functions. This imbalance remains a challenging aspect of sports activity recognition systems.

4.2 Group Activity Classification

For group activity recognition, the model achieved an accuracy of 62.75%, with a per-class average accuracy of 62.16%. The similarity between overall and per-class accuracy indicates more balanced performance across different group activities compared to individual actions.

As shown in Figure 3, the model performs best on spiking activities (l_{spike}, r_{spike}) with 73% accuracy for both. Setting activities (l_{set}, r_{set}) show moderate performance at 68% and 59% respectively. We observe that winpoint activities have the most confusion, particularly with $r_{winpoint}$ being confused with $l_{winpoint}$ 41% of the time. This suggests that distinguishing which team wins a point remains challenging, especially from the right side of the court.

The integration of self-attention mechanisms and the STGCN component helped the model capture player interactions and spatial relationships, contributing to consistent performance across various group activities despite their inherent complexity.

4.3 Rally Prediction Performance

We introduced two custom metrics to evaluate rally prediction performance:

1. **Filtered Soft Consistency Score:** This measures the alignment between neural network predictions and heuristic predictions when the heuristic is not overly confident (i.e., excluding cases where heuristic probabilities are [1,0] or [0,1]). Our model achieved a filtered soft consistency score of 73.36% with a tolerance of 0.25 across 642 qualifying clips, indicating strong agreement between learned and rule-based predictions.

2. **Group-Aligned Consistency Score:** This measures how often the predicted rally winner (left or right team) matches the implied winner from the ground truth group activity. Our model achieved a group-aligned consistency score of 73.90%, significantly outperforming the heuristic-only baseline (49.51%).

The rally probability distributions showed interesting patterns. The neural network produced more balanced probability distributions (Left: 0.506, Right: 0.494) that closely matched the ground truth distribution (Left: 0.505, Right: 0.495), while the heuristic approach showed a left-team bias (Left: 0.565, Right: 0.435). This demonstrates that our neural model successfully learned to balance the innate biases present in the rule-based heuristic approach.

4.4 Rally Winner Accuracy

When evaluating the binary classification of which team would win the rally, our neural network approach achieved 73.90% accuracy, substantially outperforming the heuristic-only approach (49.51%). This significant performance gap demonstrates the value of combining learned patterns with domain knowledge in predicting complex outcomes like rally winners in volleyball.

The superior performance of the neural approach over the heuristic method suggests that the model successfully captured subtle interactions and dynamics beyond simple rule-based logic. This is particularly impressive given the complex and fast-paced nature of volleyball rallies, where outcomes depend on numerous factors, including player positioning, momentum, and team coordination.

5 Visualization Results

We implemented a visualization module to qualitatively evaluate our system's performance on volleyball sequences. Figure 4 shows an example of our model's predictions, with players represented by bounding boxes colored according to their predicted actions.

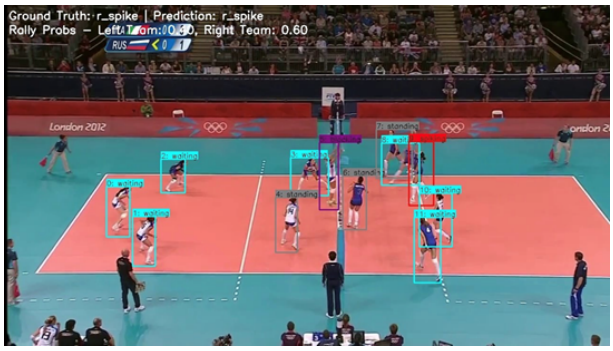


Figure 4: Model prediction showing classified player actions (spiker, blocker etc.) with group activity prediction and rally outcome probabilities

In this offensive scenario, we can observe a spiker and blocker engaged in direct interaction, with supporting players correctly classified as waiting or standing. The group activity is accurately predicted as "r_{spike}" matching the ground truth, with rally probabilities showing a clear advantage for the right team (Left: 0.40, Right: 0.60). This visualization demonstrates the model's ability to identify key players while correctly recognizing group dynamics, highlighting the effectiveness of our multi-task approach in capturing both individual actions and team-level activities.

6 Discussion and Related Work

Our volleyball activity recognition and rally prediction system demonstrates the potential of multi-task learning for understanding complex team sports dynamics. Despite the class imbalance challenges evidenced by the gap between overall (69.25%) and per-class accuracy (40.17%) for individual actions, our approach shows promising results, particularly for rally prediction where our neural network significantly outperforms the heuristic approach (73.90% vs. 49.51%).

Group activity recognition in sports videos has evolved from early approaches using handcrafted features to deep learning methods. Ibrahim et al. [3] pioneered this field with a hierarchical LSTM architecture for volleyball activity recognition, while Wu et al. [5] introduced Actor Relation Graphs for modeling player relationships. Our STGCN approach extends these ideas but specifically incorporates domain knowledge about volleyball positioning.

The confusion between left and right winpoint activities (41% error rate) suggests limitations in spatial understanding that might be addressed through ball tracking, a direction largely unexplored in current literature. Most related work in outcome prediction focuses on other sports, such as basketball [4] or foot-

ball [1], typically using trajectory data rather than visual features.

Self-attention mechanisms, as explored by Gavrilyuk et al. [2], have proven effective for capturing player dependencies, which we extend by combining with temporal LSTM modeling. The fixed proximity threshold (150 pixels) in our graph construction presents limitations for certain game situations, highlighting the need for adaptive approaches.

Our choice of AlexNet as the backbone architecture balanced computational efficiency with feature quality for our dataset size, though more modern architectures like ResNet-50 could potentially improve performance. The integration of neural networks with domain-specific heuristics allowed our system to correct for inherent biases in rule-based approaches, demonstrating the value of hybrid systems in sports analytics.

7 Conclusions and Future Work

We presented a hierarchical deep learning architecture for volleyball activity recognition and rally prediction from videos of past matches, integrating individual player action recognition, group activity classification, and rally outcome prediction in a multi-task learning framework. Our system combines a pre-trained CNN backbone with LSTM-based temporal modeling, self-attention mechanisms, and STGCN to capture volleyball gameplay dynamics.

Experimental results on the Volleyball Dataset demonstrate the effectiveness of our approach, particularly for rally prediction where our neural network-based method achieves 73.90% accuracy, significantly outperforming the heuristic-only approach (49.51%). The integration of spatial relationship modeling proved valuable for understanding team formations, while the multi-task learning framework enabled knowledge sharing between interrelated tasks.

Future work directions include:

- **Incorporating ball tracking** to provide additional context for distinguishing similar activities
- **Implementing adaptive graph construction** to better capture player relationships in different game phases
- **Extending the self-attention mechanism** to operate across temporal dimensions could allow the model to identify key moments in the sequence
- **Exploring deeper architectures** such as ResNet or Vision Transformers to enhance feature extraction capabilities

8 Team Contributions

Shiven built the model architecture, focusing on STGCN and self-attention mechanisms for spatial awareness.
Aditya developed the training pipeline, balancing various optimization techniques for multi-task learning while implementing gradient accumulation and mixed-precision training.
Ethan handled data pipeline implementation and created visualizations for player interactions.
Tanmay designed custom evaluation metrics for rally prediction and integrated system components, ensuring all parts worked together effectively.

9 Link to Code Repository

For the complete source code and documentation, click on this link: GitHub Link
or copy the below URL:
`github.com/ethanwong05/FAI_VolleyballAnalyzer`

References

[1] Javier Fernández, F. C. Barcelona, and Luke Bornn. Wide open spaces: A statistical technique for measuring space creation in professional soccer. In *MIT Sloan Sports Analytics Conference*, 2019.

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